

The recent role of Docker in the IT field

1. Docker is a tool that is designed to benefit both systems administrators and developers, making it actually a part of many DevOps (developers + operations) tool chains. For developers, it means that they can simply focus on writing code without worrying about the system that it will be running on. Docker makes it possible to set up a local development environment that is exactly like the live server. It helps to run multiple development environments, that too from the same host having unique software, configurations, operating systems and test projects, on new or different servers. It also gives them an opportunity to get a head start by using one of the thousands of programs already designed to run in Docker as a part of their application. For those in operations, Docker provides flexibility and potentially reduces the number of systems that are needed because of its small footprint and lower overhead.
2. Docker, an open source containerisation platform, isn't just the darling of several Linux powers such as Red Hat and Canonical. Various proprietary software companies such as Microsoft have also adopted it. Red Hat OpenShift 3 is one of the latest technologies that uses Docker Container and Kubernetes. OpenShift 3 has evolved and gained great momentum since its inception in 2015. It uses a Git Repository to store the source code of an application and Docker registry to manage Docker images.
3. Docker has also been used as a tooling mechanism to improve the resource management, process isolation, and security in the virtualisation of any application. This had been made clear with the launch of Project Atomic, a lightweight Linux host specifically designed to run Linux containers. The main focus of this project is to make containers easy to run, update and roll back in an environment that requires fewer resources than a typical Linux host.
4. eBay has been quite focused on incorporating Docker into its continuous integration process so as to standardise the deployment across a distributed network of different servers that run as a single cluster. The company also isolates application dependencies inside containers in order to address the issue of each server having different application dependencies, software versions, and special hardware. This means the host OS need not be the same as the container OS, and eBay's end goal is to have different software and hardware systems running as a single Mesos cluster.
5. CompileBox, a Docker-based sandbox, can run all untrusted code and return the output without any risk to the host on which the software is actually running. During the development of CompileBox, developers considered using Ideone, Chroot jails and traditional virtual machines, but Docker was selected as the best option

among all of them. Chroot does not provide the necessary level of security, whereas Ideone can quickly become cost-prohibitive, and virtual machines also take an exceptionally long time to reboot after they are compromised. Hence, Docker was the obvious choice for this application, as malicious code that attempts to destroy the system is limited to the container, and containers can be created and destroyed as quickly as needed.

Advantages of Docker

1. **Rapid application deployment:** Docker requires minimal runtime for any application, reducing the size and allowing quick deployment.
2. **Portability across machines:** An application with all its dependencies can be bundled into a single container, which is independent from the host version of the Linux kernel, deployment model or platform distribution. This container can be simply transferred to another machine that runs Docker, and executed there without any compatibility issues.
3. **Version control and component reuse:** We can track the successive versions of a Docker container, inspect the differences, or even roll back to previous versions. Docker reuses components from the preceding layers, which makes them lightweight.
4. **Sharing:** We can use a remote repository to share our Docker container with others. Red Hat even provides a registry for this purpose, and it is also possible to configure our own private repository.
5. **Lightweight footprint and minimal overhead:** Docker images are usually very small, which helps in rapid delivery and reduces the time to deploy any new application containers.
6. **Simplified maintenance:** Docker also reduces the effort and risk of problems caused due to application dependencies.
7. **Security:** From a security point of view, Docker ensures that applications that are running on containers are completely segregated from each other, granting us complete control over the traffic flow and management. No Docker container can look into processes that are running inside another container. From an architectural standpoint, each container actually gets its own set of resources, ranging from processing to network stacks. 

References

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